

14.06.2026
lec-01

~~Hasan~~
~~Raja Rashidul Hasan~~

CAD

Thursday 12-2:30
D No 209

Raja Rashidul Hasan

Room : Annex 9
level 2

Mid-Term

Attendance - 10

Report / Assignment - 40

Quiz - 20

Performance - 30

Deadline Next class

Learn

Final

Attendance - 10

Report / Assignment - 10

Performance - 30

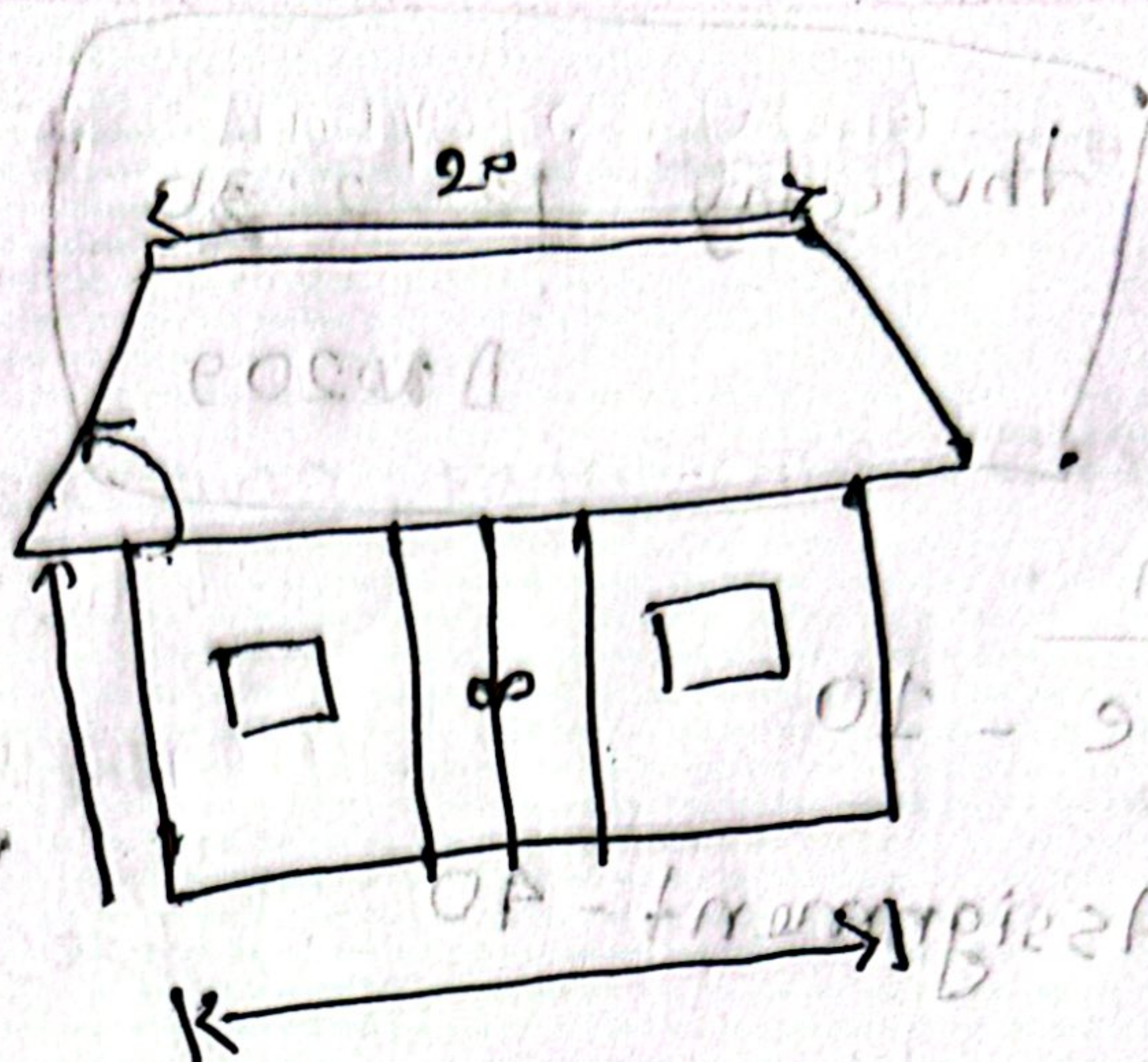
OEL Report (Group task) - 30

OEL Presentation Viva - 20

Application

Autocad software
2007

CAD - K - Serial No - Dep't - Assignment



Exp-1: Introduction to Engineering Drawing and Computer Aided Design & Drafting and Familiarization with AutoCAD software.

* Visible line

* Object Line

* Hidden line

* Center line ?

the center line



* Extension / Dimension line — Angle, Radius

* Projection → Process of identify each

Size of the object

① orthographic

②

Front view

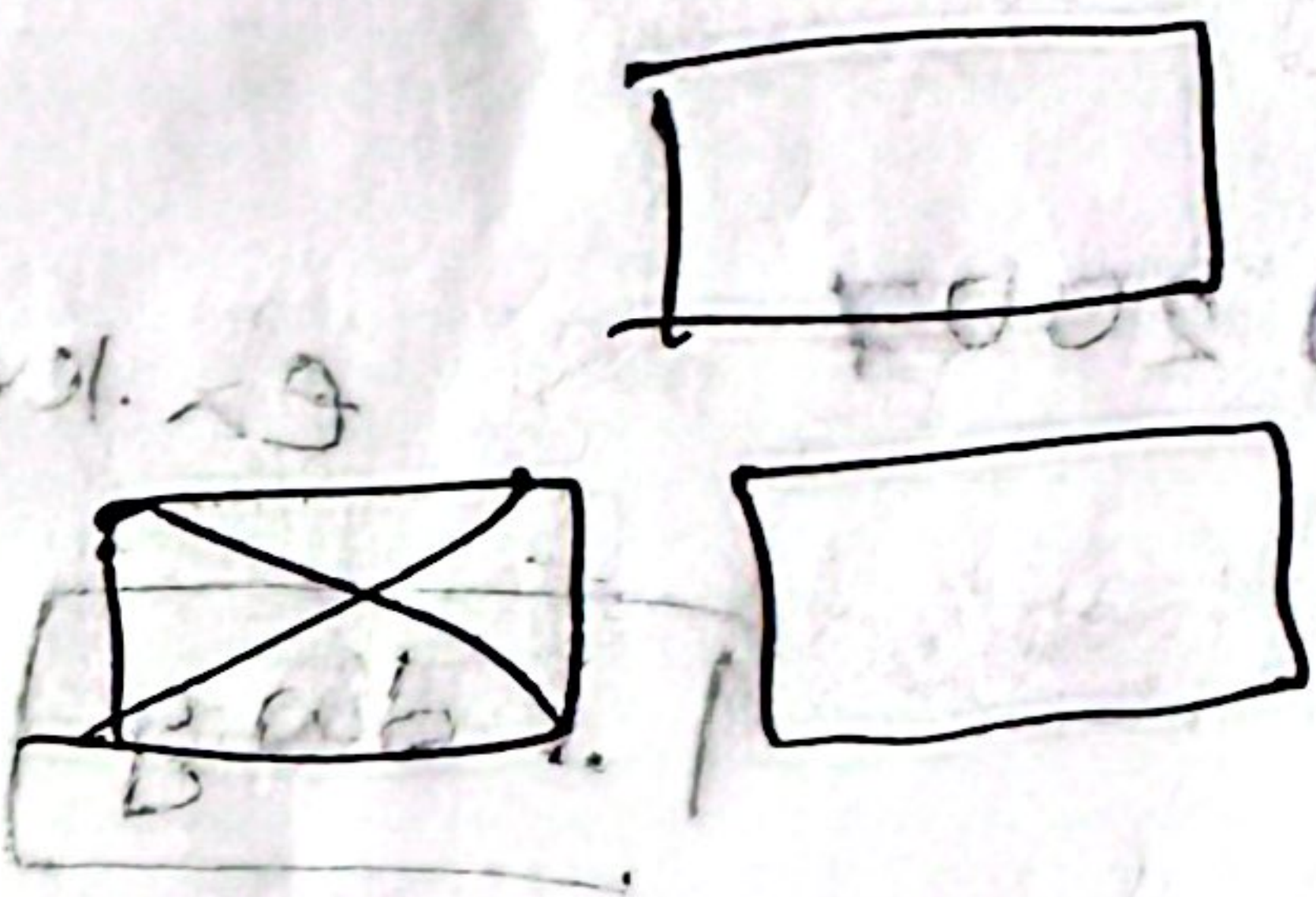
- Top
- left view

Right view

Isometric projection (3D)

length width height visible in
object

serial No: 22



Why do you CAD Design.

AutoCAD

Solid Works

engineering

CATIA

AutoDesk Inventor

TubboCAD

Why do we use area, major, used area?

- Civil
- Mechanical
- Electrical / Electronic
- Architecture
- GIS
- Multimedia

3D/Classic ✓

Title bar

AutoCAD 2007

extension

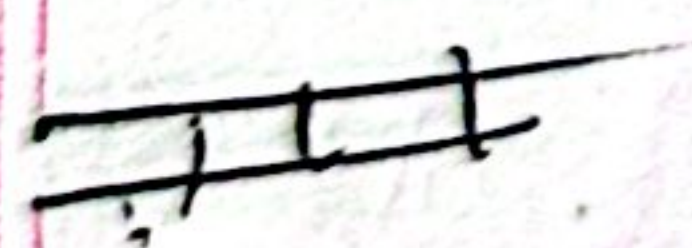
Command

1.dwg

By layer (color)

File

Status bar



ON/off

① Ortho

is used for drawing the

straight line

vertically or horizontally

OSNAP

is used to indicate in

point end, mid point,

center point etc

Setting for every new opening otherwise (3 setting)

1) You Need

Go menu -> format -> units

length type
-> engineering

Precision (2nd option)
"0.00"

(Inches)

OK

2) Format ->

Dimension style

Dimension style -> modify ->

primary units



Same as
unit & Precision

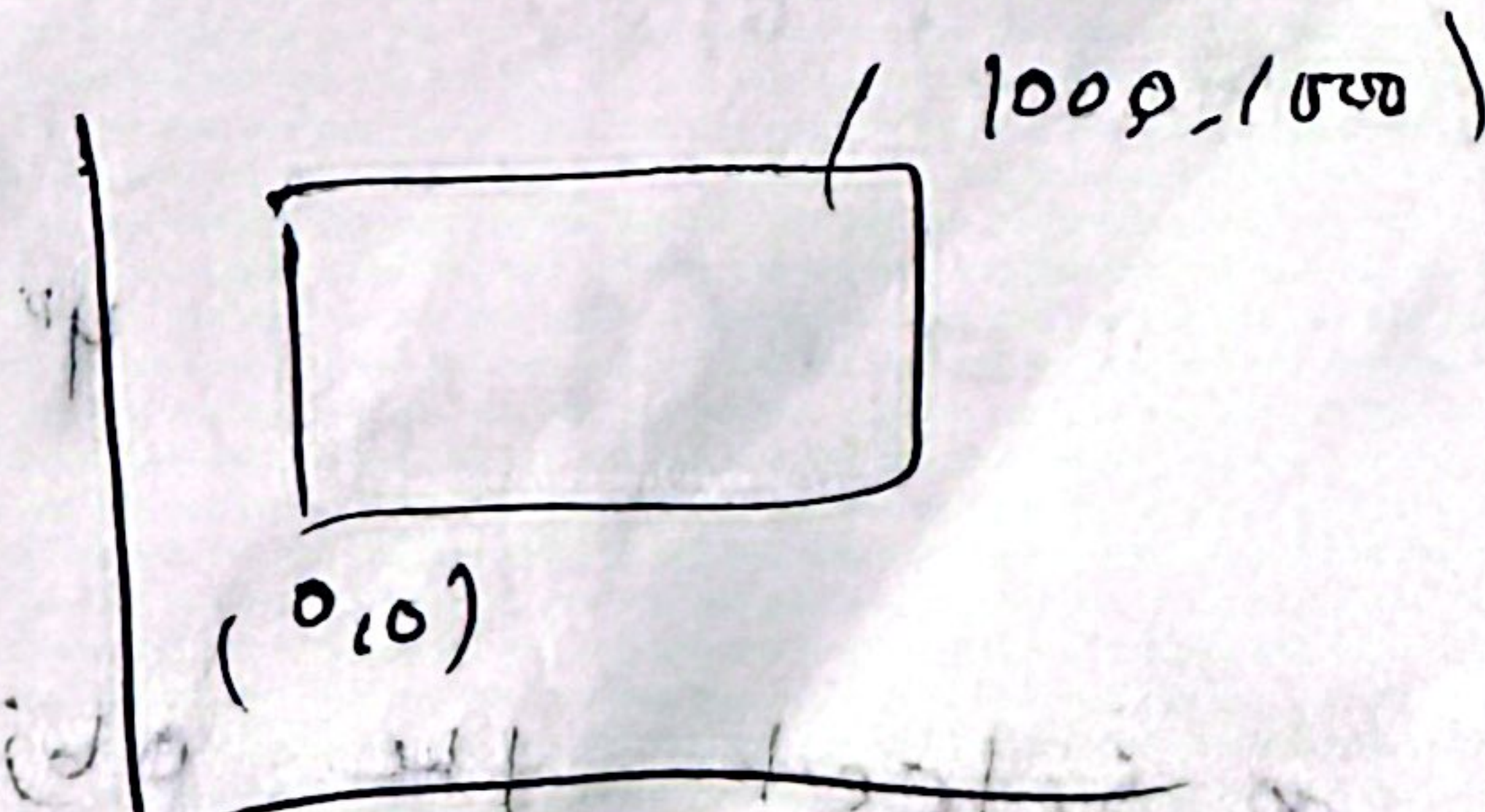
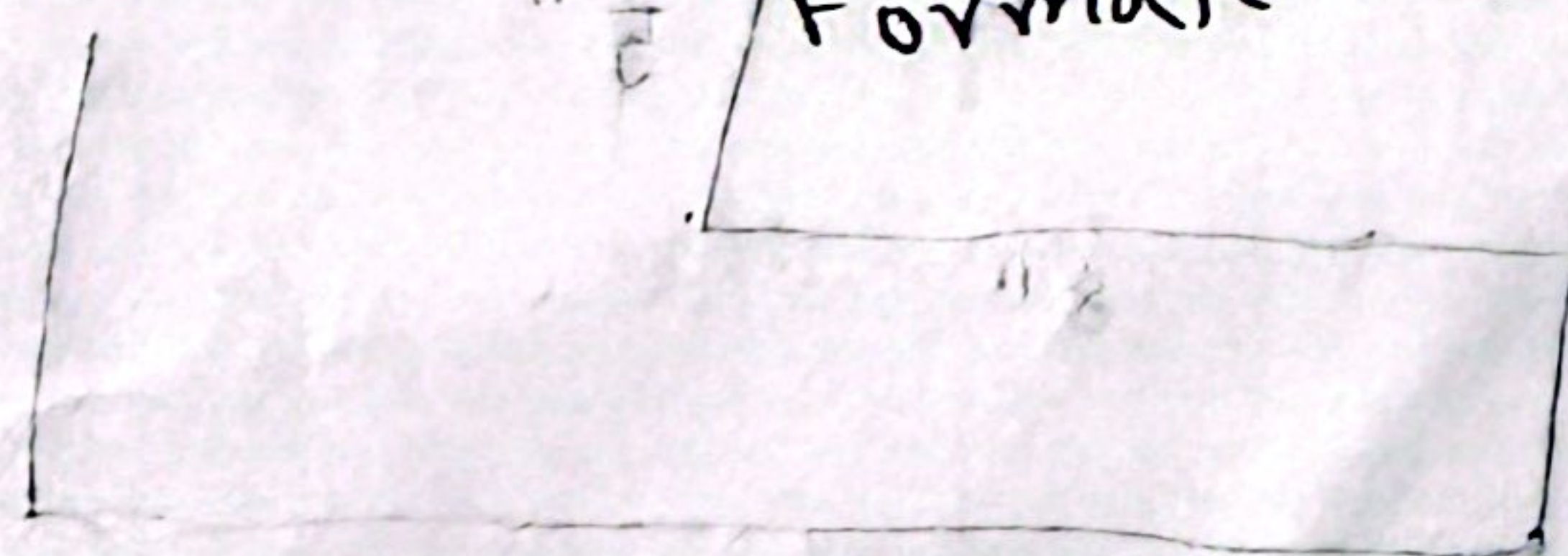
Another option

text -> text height

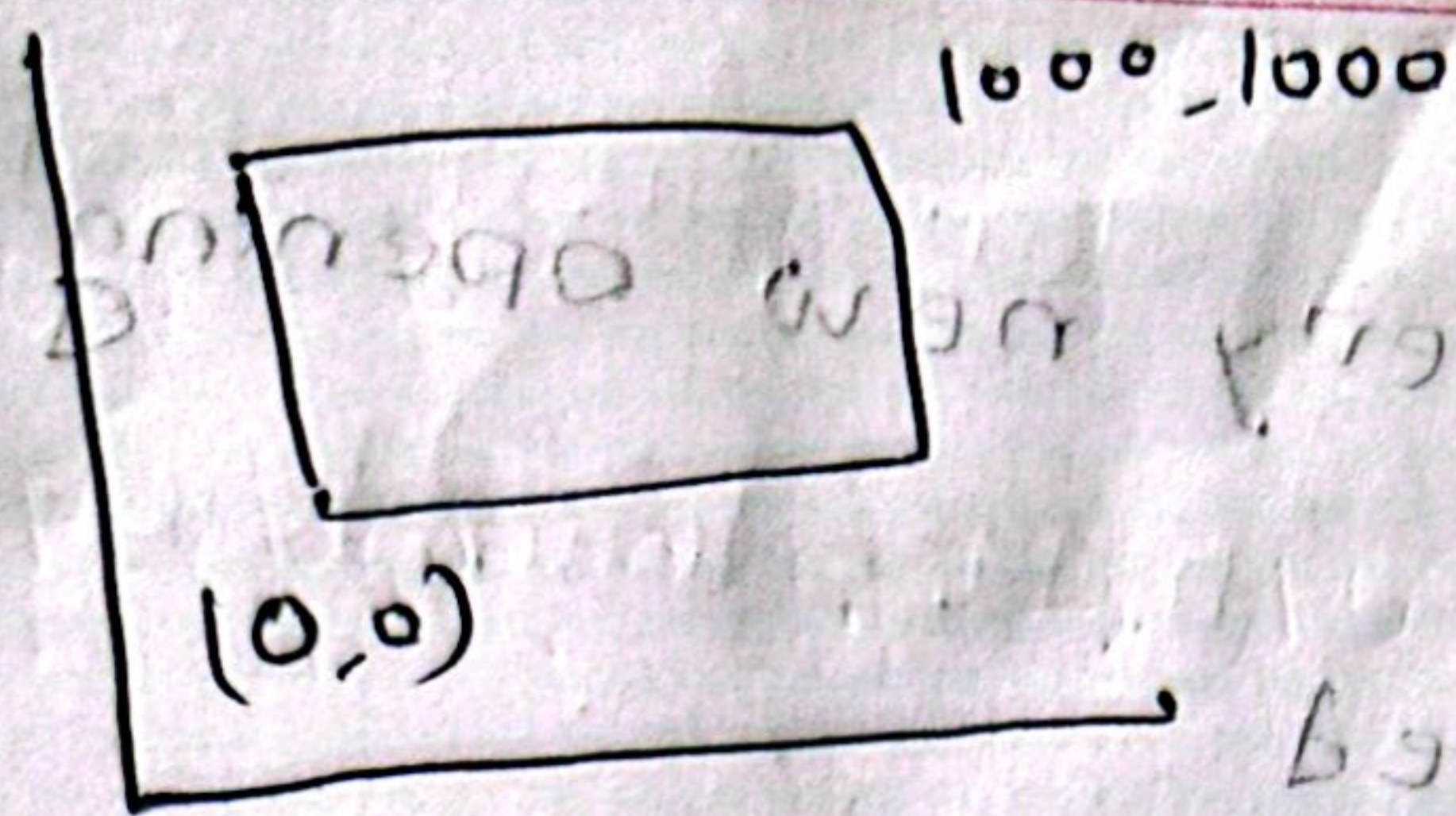
OK

3) Drawing limits

Format - Drawing limit



tools → option → color -



নিচের Command

→ line choose

Ortho

বন্ধ হওয়া জন্য Esc

* Straight line এর জন্য ortho on করে দি

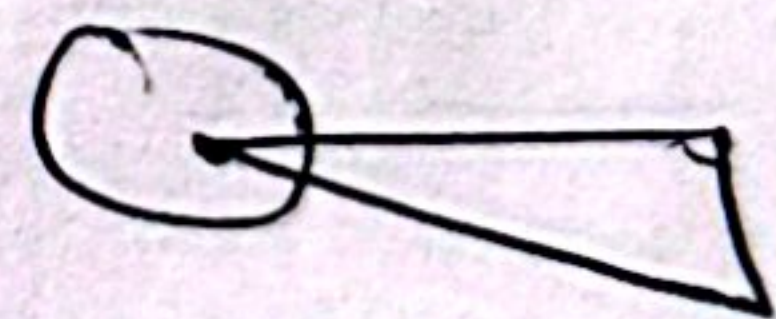
OSNAP

→ Setting → Clear all →

End, mid, center select

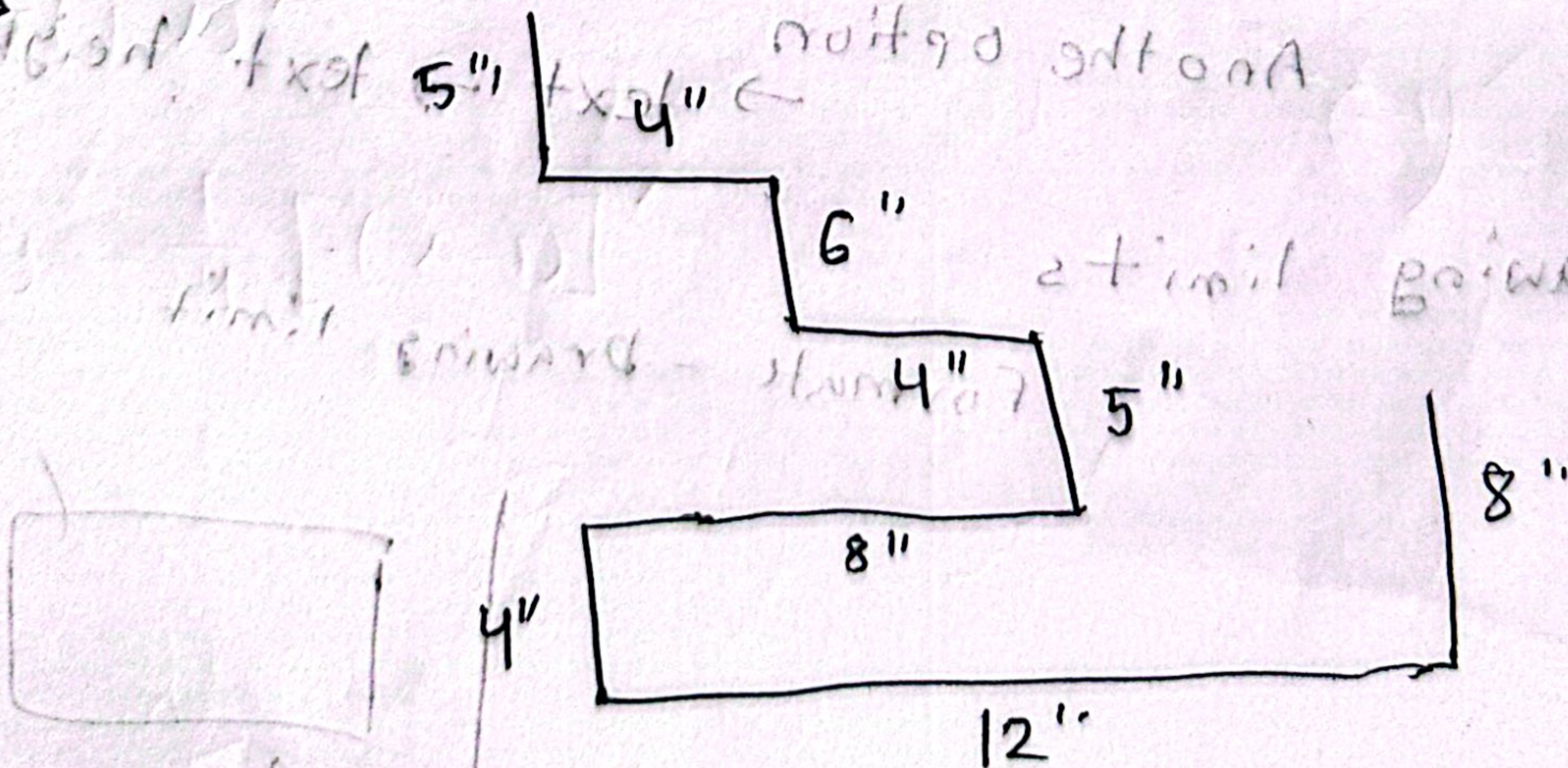
Turn off orth and connect

on থাকলে line connect করতে পারবে



XP-2,3
Orthographic
Submission
Assignment
2
Graphic Draw

Dimension → linear/Align/ → line → Esc



Select the object → Copy selection
→ move

28/06/2026
lec-03

28/06/2026
Fo-29/

CAD

click [A] → with click with how 2/3 times for [text]

* Trim :- used for cut or delete a specific portion of object

Command - tr

Isometric projection

Line → shift @ then right click

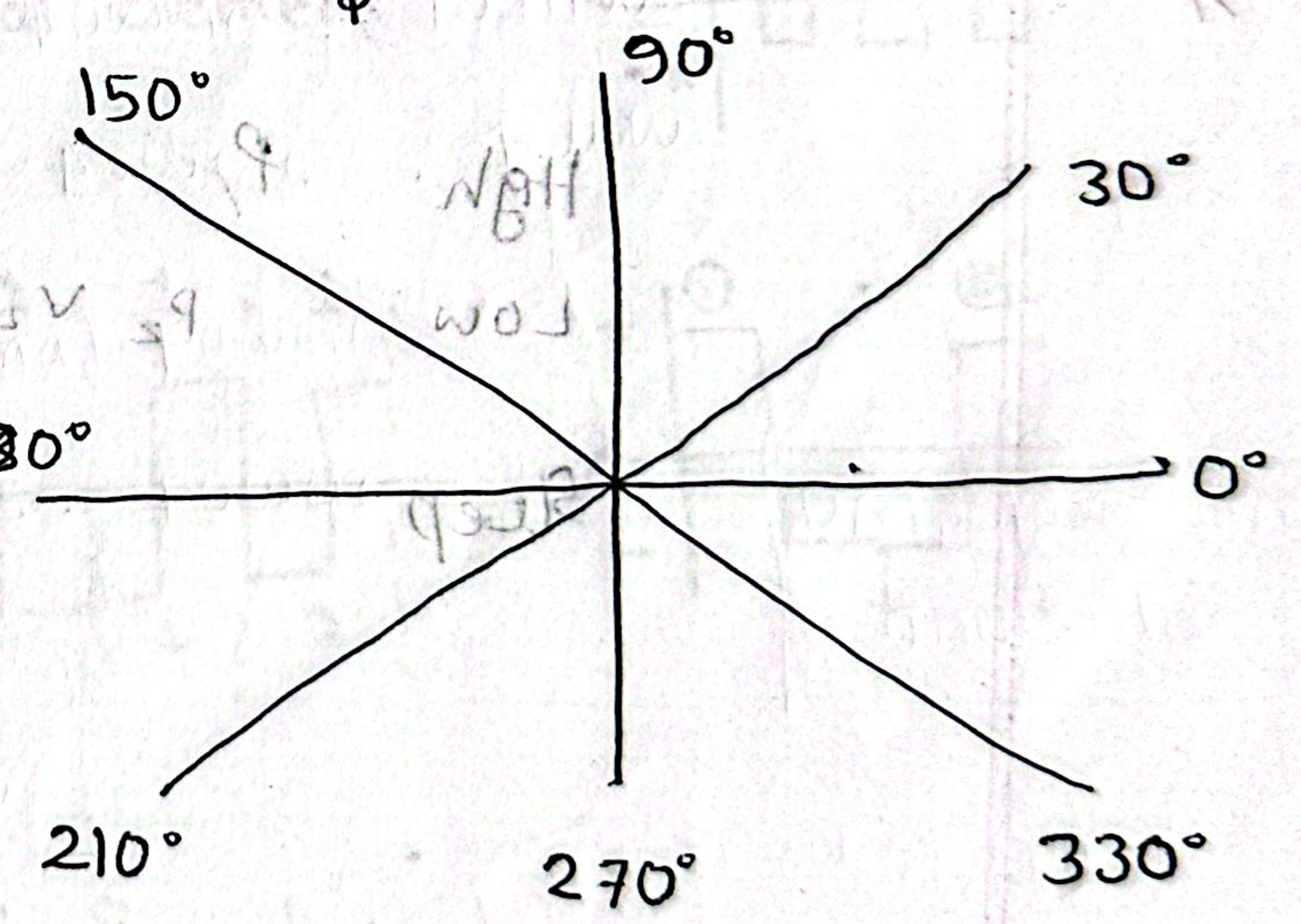
value (length)

Shift <

value (Angle)

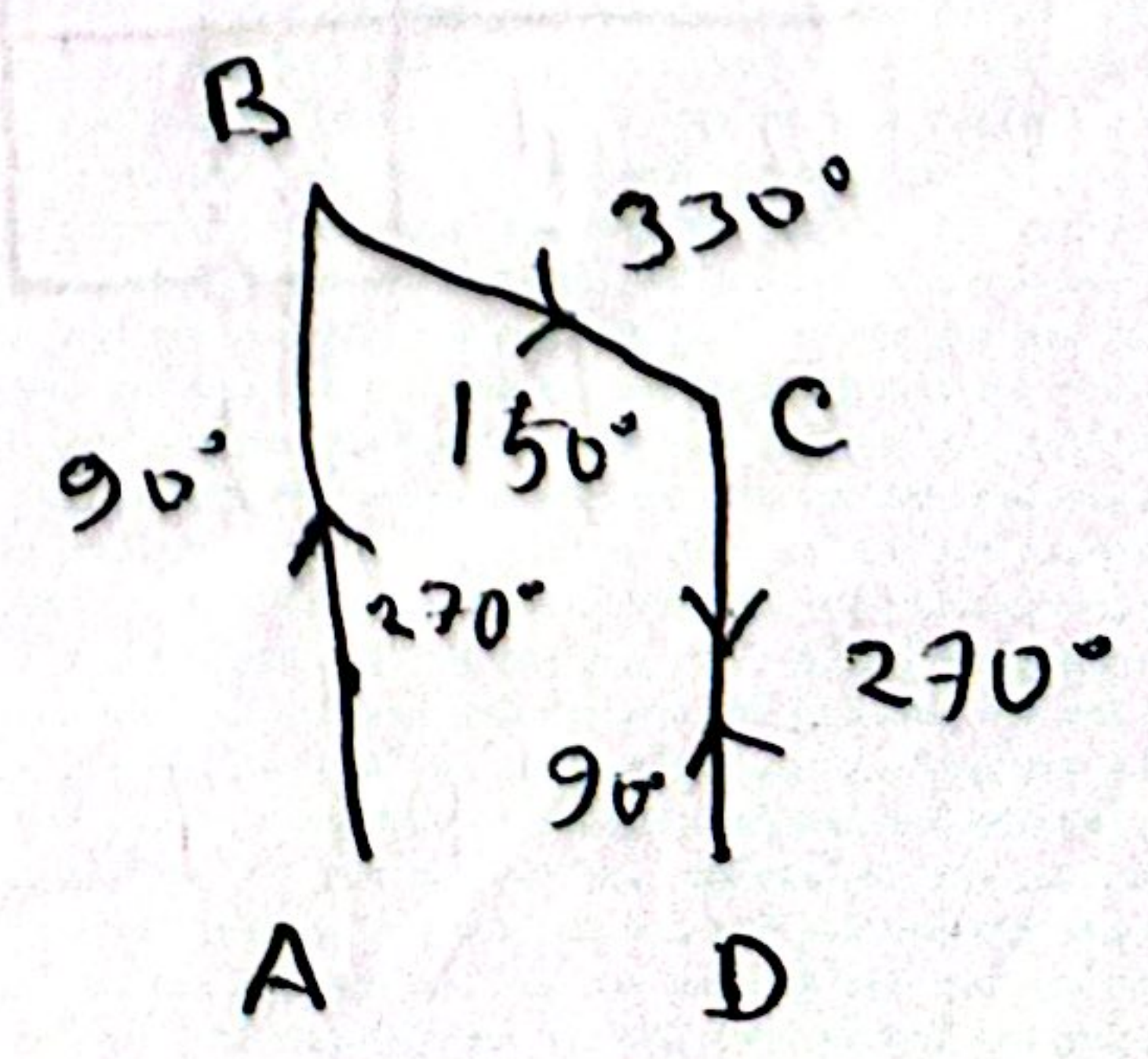
Enter

ODNAP
On the
Connect
to



Performance

Isometric
ortho



CAD Assignment M1

Summer 2025-26

Name: [blank]

ID: [blank]

Dept: [blank]

Sec: [blank]

Serial: [blank]

2/08/2026
lec-07

CAD (1 hour)

2/08/2026
lec-07

Report - 30

Performance - 30

Viva - 20

4 Report - 10 (4-1)

att - 10

Precision 0 feet 0 inch

Format $\rightarrow$ architect Length type

Format $\rightarrow$ unit $\rightarrow$ architect $\rightarrow$ mode $\rightarrow$ Precision 0'

Dimension style $\rightarrow$

③ 1/8" (1/8")

0' 8"

6' 8" $\rightarrow$ Shift $\rightarrow$ Press $\rightarrow$

Offset used to draw parallel line of

an object

line $\rightarrow$ 5' 9" $\rightarrow$ enter

Offset & dimension

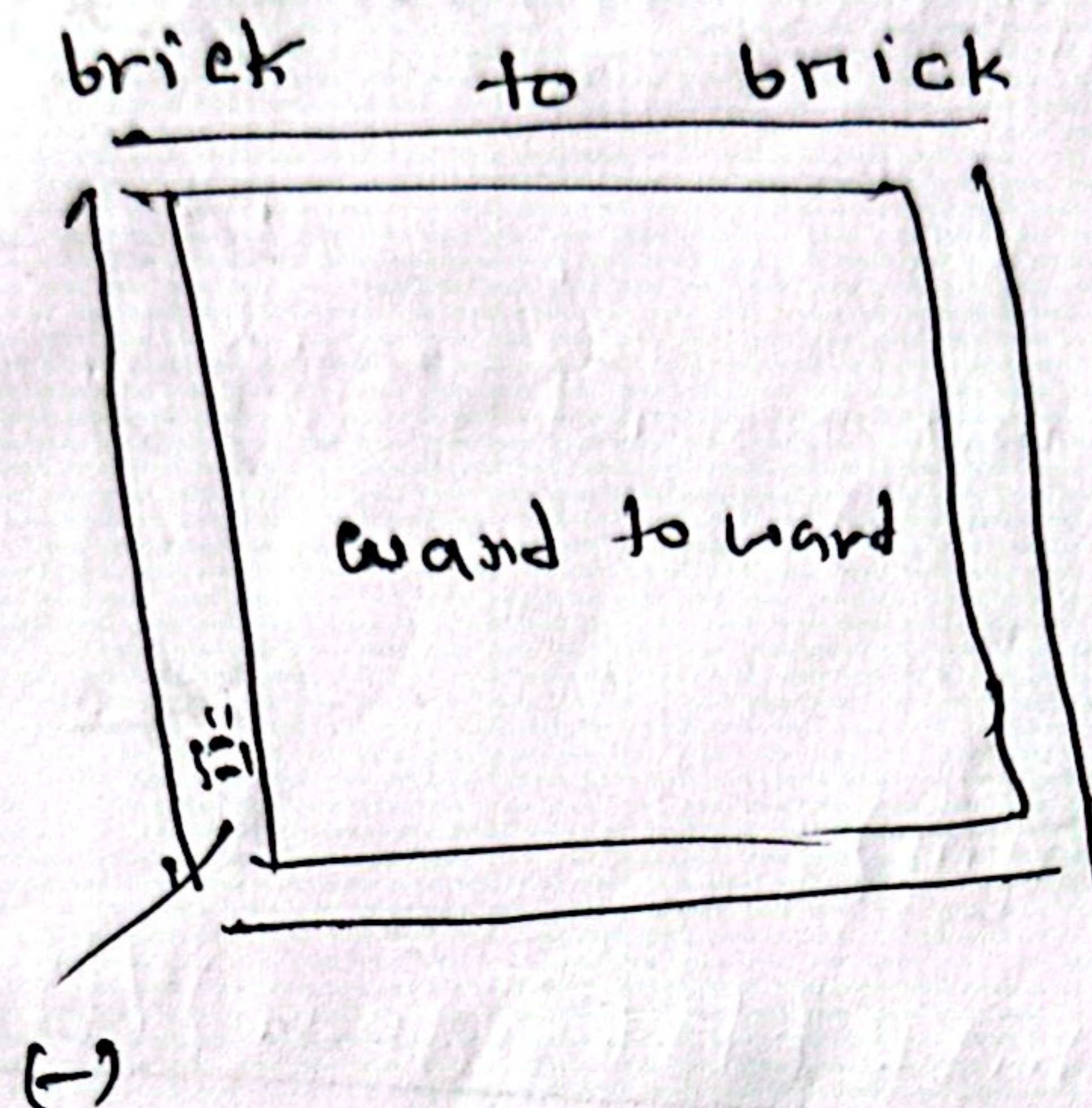
9/10/8/2026
lec-02

lab-2

CAD

8'-6" → 8 feet 6 inch
8' x 6" → 2 x 6

F1



15'-5"
-10"

14'-12"-5"
= 14'-17"
-10"
14'-7"

Start from left top corner →

move mouse and enter

turn of OSNAP → line (দ্রষ্টব্য)

LIST OF DESIGN

→ short form

Fan → F → 

Light → L → 

Switch Box → Rectangle 

Switch socket → SS → 

starting level → SL → 

18 feet or 20 feet 2 fan

1 → tube

2 → Normal

बादला light Ceiling → 

SS → multisockets

3 phase - 3 pole socket AC, microwave

Ab → Antenna junction board

30/08 OEL Viva

20 mark

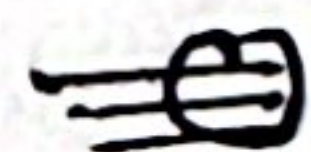
Group

- 192

Numbering

बादला

light



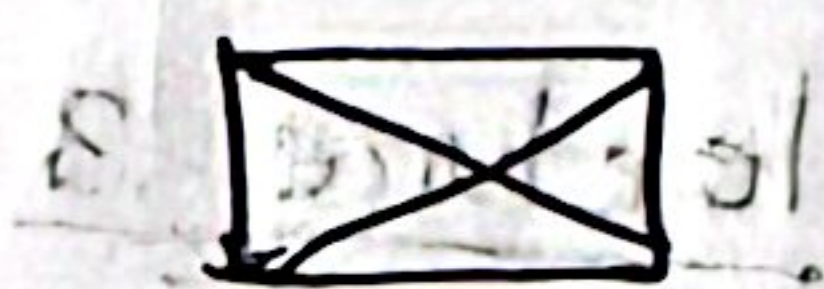
15A

3 phase

3 pole

3 pole

SUB distribution board SDB



Circuit breaker

Each apartment 1 sub distribution board
building main distribution board

Line voltage

Circuit breaker → For protection

overload & short circuit

[Name → short Name → Symbol]

Input / output

Input / output

Binary form

4 x 4 MUX

Binary 2's complement (not used)

Left right - table given

Splitter → split

Register 2 store 2M data

Which Distribution

Input Register
MUX 2 bits

Input Register

Logic unit
which performs the

31/08/2026
lec-06

Data Communication

Network Topology and switching :-

* Timing diagram → Draw → exam

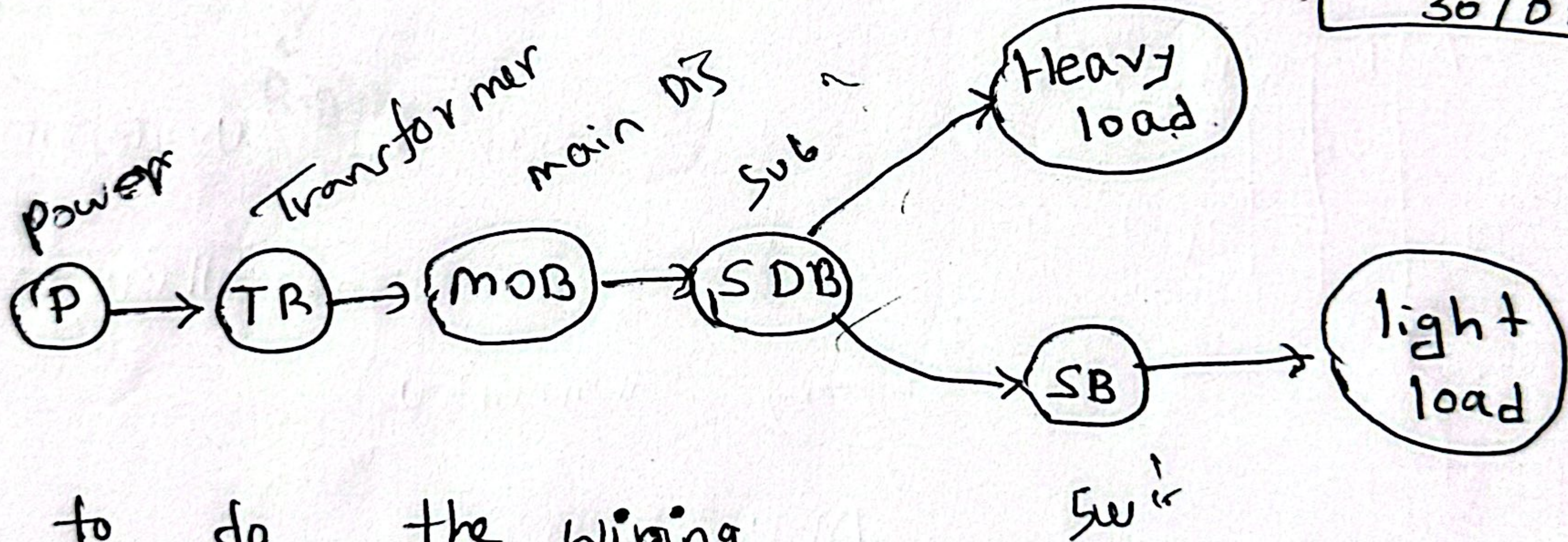
Component of communication situation → map

Assignment

Submit-06 file

CAD

Performance test
Next Sunday
30/08/26



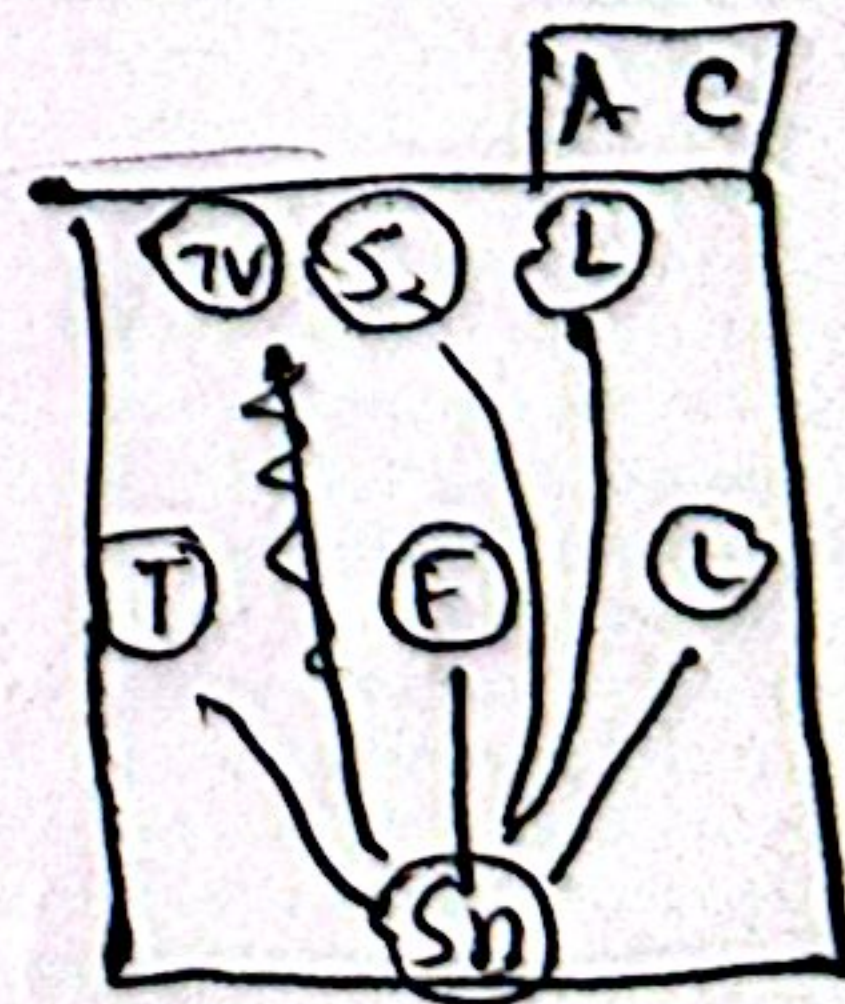
How to do the wiring

Step-0) Replace all symbols by letter

step-1) Connect all light loads to the dedicated specified switch board.

Step-2) Connect all heavy loads directly to the sub distribution board, through separate circuit breaker

Step-3) Connect all switch board to the sub distr

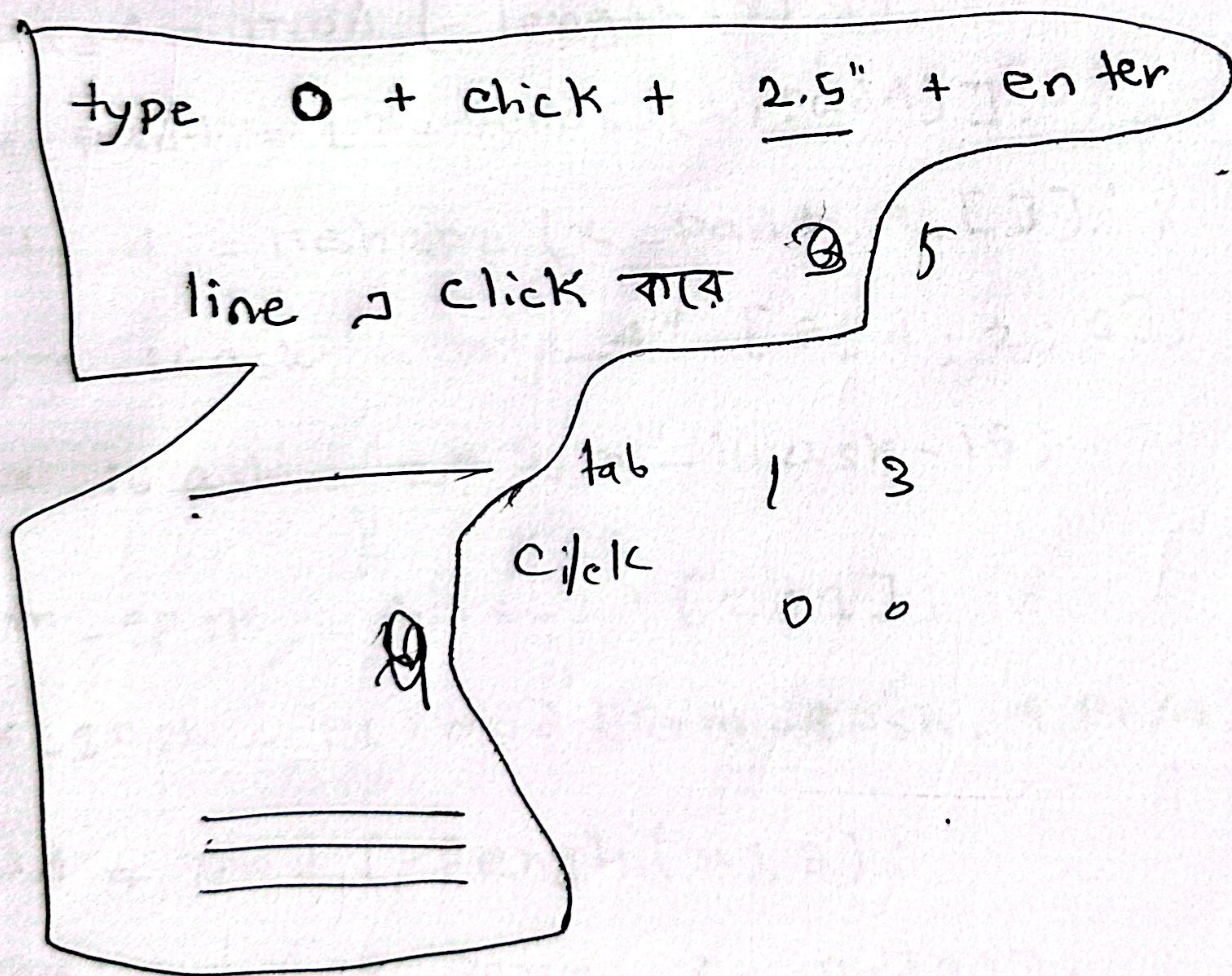


board. through the circuit breaker. If needed
do the balancing

step-4) connect all telephone line to the TJB
and TV line to AIB

5) Connect sub distribution board to the main
Distribution board through a PVC Riser

unit
SS



15-12
20-15